Kernelized Burer–Monteiro Dynamics: A Spectral Theory of Implicit Bias for Neural Metric Learning

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Abstract

We show that gradient flow on a deep ReLU embedding network, trained with a twice-differentiable loss on its Gram matrix $G = FF^\top$, evolves under a *kernel-preconditioned Burer–Monteiro flow* $\dot{G} = -2\,(KEG + GEK)$, where $E = \partial L/\partial G$ and K is the depth-L neural tangent kernel. When K is the identity this recovers the classical Burer–Monteiro dynamics for low-rank semidefinite optimization. Beyond this equivalence, the kernel form yields a *spectral implicit bias*: in the eigenbasis of K, each mode of G evolves at a rate set by the corresponding kernel eigenvalue, so structure that lives in K's leading eigenspace is fit quickly while structure in its trailing eigenspace is essentially frozen in the NTK regime. This is a testable, non-trivial prediction that does not follow from the NTK ODE without the BM viewpoint. We verify both the equivalence (multi-seed, $N \in \{10, 30, 100\}$, depth $L \in \{1, 2, 3\}$) and the spectral-bias prediction (synthetic and Fashion-MNIST) and show that the predicted mode-wise rate ordering holds across regimes.

1 Introduction

Neural metric learning trains an embedding $f_\theta : \mathbb{R}^d \to \mathbb{R}^k$ so that $\|f_\theta(x) - f_\theta(y)\|^2$ encodes a desired similarity geometry. Two theoretical frameworks bear on this setting from very different angles. The *neural tangent kernel* [Jacot et al., 2018, Lee et al., 2019] describes sufficiently wide networks as kernel methods governed by a fixed kernel Θ, and yields linear ODEs for the network's predictions under gradient flow. The *Burer–Monteiro factorization* [Burer and Monteiro, 2003, 2005], central to low-rank semidefinite programming, parameterizes a positive semidefinite matrix X as $X = UU^\top$ and uses U as the optimization variable; the resulting gradient flow on X takes the form $\dot{X} = -2(EX + XE)$.

The two frameworks are usually treated in isolation. We connect them by observing that an embedding network with twice-differentiable loss on $G = FF^\top$ is a Burer–Monteiro factorization in non-linear feature space, and that the NTK enters as an explicit *kernel preconditioner* on the BM dynamics. Concretely, in the NTK regime the network's Gram matrix evolves according to

$$\dot{G} \;=\; -2\left(K\,E\,G + G\,E\,K\right), \qquad E = \partial L/\partial G, \tag{1}$$

which reduces to the classical BM flow when $K = I$. We call (1) the *kernelized Burer–Monteiro* (kBM) flow.

The equivalence is itself a useful conceptual bridge, but on its own it does not say more than the standard NTK ODE in different coordinates. The contribution of this paper is to use the kBM viewpoint to extract a *spectral implicit bias* that does not follow from NTK alone. Diagonalizing $K = U\Lambda U^\top$ and writing $\widetilde{G} := U^\top G U$, $\widetilde{E} := U^\top E U$, the kBM flow becomes

$$\dot{\widetilde{G}} \;=\; -2\,(\Lambda\,\widetilde{E}\,\widetilde{G} + \widetilde{G}\,\widetilde{E}\,\Lambda),$$

so the time derivative of mode (i, j) is multiplied by $\lambda_i + \lambda_j$. Modes corresponding to small kernel eigenvalues are essentially frozen. This makes a sharp, testable prediction: the trajectory of $G(t)$ confines its motion to K's top eigenspaces, regardless of where the loss landscape would otherwise prefer it to go.

Contributions.

1. **Equivalence theorem.** Under gradient flow on a single-layer ReLU network with the standard NTK approximation, the Gram-matrix dynamics take the kBM form (1), recovering the classical BM dynamics when $K = I$ (Theorem 2).

2. **Depth-L extension.** The same kBM equation governs a depth-L fully-connected ReLU network, with K replaced by the depth-L NTK from the recursive Jacot construction (Theorem 4).

3. **Spectral implicit-bias theorems.** In the eigenbasis of K the kBM flow's (i, j) component evolves at rate $\lambda_i + \lambda_j$; under a benign-loss assumption this implies a model-free linear-in-λ time-constant bound (Theorem 6). Within the NTK regime, where G_0 and K share approximate eigenstructure, the bound tightens to a quadratic-in-λ prediction (Theorem 9), which matches the empirical slope ≈ 2 across configurations.

4. **Numerical illustration.** We verify the equivalence and the spectral-bias prediction with multi-seed experiments on synthetic data ($N \in \{10, 30, 100, 1000\}$, depth $L \in \{1, 2, 3\}$) and on Fashion-MNIST triplets. Empirical power-law slopes range from 1.38 to 2.19 across configurations, all consistent with the bound ≤ 2 from Theorem 9.

5. **Algorithmic payoff: spectral preconditioner.** Applying K^{-1} to the embedding-side gradient flattens the spectral bias from slope 1.99 ± 0.11 (vanilla) to 0.74 ± 0.10 (preconditioned), a one-line modification to backpropagation derived directly from the kBM viewpoint (Section 4.6). The same effect appears on real CIFAR-10 features.

6. **Effective hypothesis class.** As a direct corollary (Corollary 11), the kBM theory yields a quantitative effective rank of the hypothesis class on bounded training horizons: only metric structure supported on K's top-$r(T, \delta)$ eigenspace is learnable.

We are explicit about scope. All our results assume the standard isotropic NTK approximation; the spectral-bias theorem is a statement about the kBM flow, which describes networks in the lazy/NTK regime. We do not analyze feature-learning corrections, nor do we claim a generalization bound. We discuss limitations in Section 5.

2 Related Work

NTK theory. The NTK framework of Jacot et al. [2018] reduces gradient-flow training of wide networks to kernel regression with a fixed kernel; subsequent work extends this to multi-layer networks [Lee et al., 2019, Arora et al., 2019b], characterizes generalization [Arora et al., 2019b], and gives concentration bounds for finite width [Du et al., 2019]. Chizat et al. [2019] clarifies the lazy-vs-rich training dichotomy. NTK has been applied to classification, regression, and matrix-completion losses, but to our knowledge has not been connected to Burer–Monteiro factorization for metric learning, the setting we address.

Burer–Monteiro and SDP factorization. Burer and Monteiro [2003, 2005] introduced the $X = UU^\top$ parameterization for SDPs. Boumal et al. [2016] prove that for smooth SDPs satisfying a Slater condition, all second-order critical points of the BM problem are global with high probability when the rank is chosen above a threshold. Bhojanapalli et al. [2016] and Ge et al. [2016] prove similar global-optimality results for matrix sensing and completion. Park et al. [2017] extends the BM landscape analysis to non-square matrix sensing. Our equivalence is closest in spirit to these results, but in the non-linear feature space of a neural network and with a kernel-preconditioned dynamics rather than vanilla BM gradient flow.

Implicit bias of gradient descent. Gunasekar et al. [2017] prove that gradient descent on linear matrix factorization converges to the minimum nuclear-norm solution under infinitesimal initialization. Arora et al. [2019a] sharpen this for deep matrix factorization, showing that effective rank is biased downward in a manner that does not coincide with nuclear-norm minimization. Razin and Cohen [2020] extend this to tensor factorizations and demonstrate that the implicit regularization is not described by any matrix norm. Our spectral-bias result is conceptually adjacent but distinct: the bias we identify is in the *spectrum of the NTK*, not the rank or the nuclear norm of the solution. The two views are complementary; we discuss their relationship in Section 5.

Implicit bias and margin. Soudry et al. [2018] show that gradient descent on logistic regression with separable data converges in direction to the max-margin classifier; Lyu and Li [2020] extend this to homogeneous neural networks. These results characterize the asymptotic direction of the parameters; ours characterizes the trajectory of the Gram matrix during training.

Metric learning. Schroff et al. [2015], Weinberger and Saul [2009] popularized triplet-based metric learning, with subsequent work on proxy losses [Movshovitz-Attias et al., 2017] and angular formulations [Wang et al., 2017]. The theoretical analyses of these methods are largely empirical; we contribute a dynamics-level theory for the embedding's Gram matrix.

Neural networks as convex regularizers. Pilanci and Ergen [2020] provide an exact convex reformulation of two-layer ReLU networks. Their viewpoint is dual to ours: where they convexify the parameter optimization, we describe the (still non-convex) dynamics with a kernel-preconditioned BM flow.

3 Theoretical Framework

3.1 Setup

Let $X \in \mathbb{R}^{N\times d}$ be a data matrix with rows x_i. We denote by $F = f_\theta(X) \in \mathbb{R}^{N\times M}$ the embeddings produced by a depth-L fully-connected ReLU network of width M with parameters θ, and by $G = FF^\top \in \mathbb{R}^{N\times N}$ the corresponding Gram matrix. We consider any twice-differentiable loss $L : \mathbb{R}^{N\times N} \to \mathbb{R}$ on G; triplet loss is the running example. Write $E := \partial L/\partial G$.

We use the standard NTK parameterization with weights initialized $\mathcal{N}(0, 1/d_{\text{in}})$ at each layer and no output normalization; under this convention G_{ij} is $\Theta(1)$ at initialization and our expressions are scale-free.

3.2 Single-layer kBM equivalence

Definition 1 (NTK at the data)**.** *The Jacobian of the embedding is $J(x;\theta) = \nabla_\theta f_\theta(x)$. The (output-summed) NTK at x_i, x_j is $\kappa(x_i, x_j) := \operatorname{tr}\big(J(x_i)J(x_j)^\top\big)/M$. We write $K_{ij} := \kappa(x_i, x_j)$ for the empirical kernel matrix.*

Theorem 2 (kBM equivalence, single layer)**.** *For a single-layer ReLU network $F_{i\cdot} = \mathrm{ReLU}(x_i^\top W)$ with $W \in \mathbb{R}^{d\times M}$ trained by gradient flow on $L(G)$, the Gram matrix G evolves, in the limit $M \to \infty$, as*

$$\dot{G} \;=\; -2\left(K\,E\,G + G\,E\,K\right).$$

For finite width M the same equation holds with the empirical kernel $K^{(M)}$ in place of K; the gap $\left\|K^{(M)} - K\right\|_{op} = O_p(M^{-1/2})$ under standard assumptions [Jacot et al., 2018, Arora et al., 2019b].

Proof sketch. Under gradient flow $\dot{\theta} = -\nabla_\theta L$, we have $\dot{G}_{ij} = \langle \nabla_\theta G_{ij}, \dot{\theta}\rangle = -\sum_{k,l} E_{kl}\,\Theta^G_{(ij),(kl)}$ where $\Theta^G_{(ij),(kl)} := \langle \nabla_\theta G_{ij}, \nabla_\theta G_{kl}\rangle$ is the Gram-Gram NTK. A direct computation using $G_{ij} = \sum_m F_{im}F_{jm}$ and the chain rule gives, in the $M \to \infty$ limit,

$$\Theta^G_{(ij),(kl)} \;=\; \kappa_{ik}G_{jl} + \kappa_{il}G_{jk} + \kappa_{jk}G_{il} + \kappa_{jl}G_{ik}.$$

Substituting and using $E = E^\top$ collapses the four terms into $KEG + GEK$. The full derivation appears in Appendix A. □

Remark 3. *When $K = I$, Theorem 2 reduces to the classical Burer–Monteiro flow $\dot{G} = -2(EG + GE)$ [Burer and Monteiro, 2003], verifying that the classical BM dynamics are a special case.*

3.3 Depth-L extension

The depth-L NTK admits the recursive form of Jacot et al. [2018]: $\Theta^{(\ell)} = \Sigma^{(\ell)} + \dot{\Sigma}^{(\ell)} \cdot \Theta^{(\ell-1)}$ with $\Theta^{(1)} = \Sigma^{(1)} + \dot{\Sigma}^{(1)} \cdot \langle x, y\rangle$, where $\Sigma^{(\ell)}$ is the layer-ℓ activation kernel (the Cho–Saul arc-cosine kernel) and $\dot{\Sigma}^{(\ell)}$ its derivative [Cho and Saul, 2009].

Theorem 4 (kBM equivalence, depth L)**.** *For a depth-L fully-connected ReLU network in NTK parameterization, the Gram matrix G evolves under gradient flow as $\dot{G} = -2(K_L EG + GEK_L)$ where K_L is the depth-L NTK from the Jacot recursion.*

Proof sketch. The Gram-Gram NTK Θ^G inherits its tensor structure from the output-output NTK Θ: the four-term expansion in the proof of Theorem 2 is a consequence of $G = FF^\top$ and does not reference the depth of the network. Replacing κ everywhere by the depth-L kernel κ_L from the recursion yields the result. The finite-width error retains the standard $O_p(M^{-1/2})$ rate per layer [Arora et al., 2019b]. □

3.4 Spectral implicit bias

The kBM flow takes a particularly clean form in K's eigenbasis, and this form yields the paper's central new result.

Lemma 5 (kBM in K's eigenbasis)**.** *Let $K = U\Lambda U^\top$ with $\Lambda = \mathrm{diag}(\lambda_1, \ldots, \lambda_N)$, $\lambda_1 \geq \cdots \geq \lambda_N \geq 0$. Define $\widetilde{G} := U^\top G U$ and $\widetilde{E} := U^\top E U$. Then the kBM flow becomes*

$$\dot{\widetilde{G}} \;=\; -2\left(\Lambda\,\widetilde{E}\,\widetilde{G} + \widetilde{G}\,\widetilde{E}\,\Lambda\right),$$

and the (i,j) entry satisfies $\dot{\widetilde{G}}_{ij} = -2\big(\lambda_i(\widetilde{E}\widetilde{G})_{ij} + \lambda_j(\widetilde{G}\widetilde{E})_{ij}\big)$.

Proof. Direct substitution of $K = U\Lambda U^\top$ and conjugation by $U^\top, U$. □

Theorem 6 (Spectral implicit bias)**.** *Let $G(t)$ solve the kBM ODE $\dot{G} = -2(KEG + GEK)$ with $L \in C^2$ and gradients bounded over the trajectory. Suppose $\lambda_i = 0$ for some index i. Then $\widetilde{G}_{ij}(t) = \widetilde{G}_{ij}(0)$ for all t and all j. More generally, if $\lambda_i \leq \varepsilon$, then over any interval $[0,T]$ on which $\big\|\widetilde{E}\widetilde{G}\big\|_{op}, \big\|\widetilde{G}\widetilde{E}\big\|_{op}$ are uniformly bounded by C,*

$$\big|\widetilde{G}_{ij}(T) - \widetilde{G}_{ij}(0)\big| \;\leq\; 2T(\lambda_i + \lambda_j)\,C.$$

Proof. The exact-zero case follows from Lemma 5: the row i of $\Lambda\widetilde{E}\widetilde{G}$ vanishes (premultiplication by a row of Λ with $\lambda_i = 0$), and the column i of $\widetilde{G}\widetilde{E}\Lambda$ vanishes. Symmetric argument for j. The bound follows by integrating $|\dot{\widetilde{G}}_{ij}| \leq 2(\lambda_i + \lambda_j)C$ over $[0,T]$. □

Corollary 7 (Mode-wise time constants)**.** *For diagonal entries, in the linearization around any point where $\widetilde{E}\widetilde{G}$ has bounded operator norm, the mode-i time constant satisfies $\tau_i \propto 1/\lambda_i$. Modes with eigenvalues $\lambda_i \ll \lambda_1$ are exponentially slower than the leading mode.*

3.5 From a linear bound to a quadratic prediction

The bound in Theorem 6 is linear in λ_i. In our experiments (Section 4.3) the empirical scaling of diagonal displacements $|\widetilde{G}_{ii}(T) - \widetilde{G}_{ii}(0)|$ is consistently *quadratic* in λ_i across depths and across both synthetic and Fashion-MNIST data. The gap between linear bound and quadratic observation is closed by exploiting a structural property of NTK initialization: G_0 and K share approximate eigenstructure, so $\widetilde{G}_0$ is approximately diagonal in K's eigenbasis with diagonal entries proportional to the kernel spectrum.

Proposition 8 (Approximate eigen-alignment at NTK init)**.** *For a single-layer ReLU network in NTK parameterization with width M and inputs drawn iid from a rotationally-invariant distribution on the unit sphere, $G_0 = F_0 F_0^\top$ converges as $M \to \infty$ to the layer-1 arc-cosine kernel $\Sigma^{(1)}$ [Cho and Saul, 2009]. Both $\Sigma^{(1)}$ and the NTK $K = \Sigma^{(1)} + \dot{\Sigma}^{(1)} \odot \langle x_i, x_j \rangle$ are functions of the input correlation matrix $\langle x_i, x_j \rangle$ alone; under rotational invariance they share an asymptotic eigenbasis (the spherical harmonics on S^{d-1}). Consequently, $\widetilde{G}_0 := U^\top G_0 U$ is approximately diagonal, and the diagonal entries satisfy $\widetilde{G}_0[i,i] = c_i \lambda_i$ with c_i bounded and slowly varying across the spectrum.*

Proof. Both $\Sigma^{(1)}$ and K are scalar functions of $\langle x_i, x_j \rangle$ on the unit sphere, so the Funk–Hecke theorem [Atkinson and Han, 2012, Theorem 2.22] diagonalizes them in the spherical-harmonic basis with eigenvalues given by Gegenbauer integrals of κ_1 and $\kappa_1 + \kappa_0(\cdot)\cdot t$. The ratio $c_l = \lambda_l(\kappa_1)/\lambda_l(\kappa_1 + \kappa_0 \cdot t)$ lies in $(0,1)$ and is bounded away from 0 uniformly in l, using the asymptotic decay rates from Bach [2017]. Empirical kernels on N samples concentrate around the population kernels at rate $O(N^{-1/2})$ [Tropp, 2015]; eigenvectors align via Davis–Kahan. Full details in Appendix A.4. □

Theorem 9 (Refined spectral implicit bias)**.** *Let $G(t)$ solve the kBM ODE $\dot{G} = -2(KEG + GEK)$ with $L \in C^2$. Assume:*

(A1) $\widetilde{G}_0$ is diagonal in K's eigenbasis with diagonal $\widetilde{G}_0[i,i] = c_i\lambda_i$, $c_i \in [c_{\min}, c_{\max}]$;

(A2) $\widetilde{E}(s)$ has uniformly bounded entries $|\widetilde{E}[i,j](s)| \le E$ for $s \in [0,T]$.

Then, to leading order in T,

$$\widetilde{G}_{ii}(T) - \widetilde{G}_{ii}(0) \;=\; -\,4\,T\,\lambda_i\,(\widetilde{E}_0\,\widetilde{G}_0)_{ii} \;+\; O(T^2),$$

and under the additional assumption (A3) that $\widetilde{E}_0\,\widetilde{G}_0$ is approximately diagonal at initialization in the sense $|(\widetilde{E}_0\,\widetilde{G}_0)[i,i]| \le c_{\max}\lambda_i\,|\widetilde{E}_0[i,i]| + O(\sqrt{N}/M^{1/2})$, we have

$$\boxed{\left|\widetilde{G}_{ii}(T) - \widetilde{G}_{ii}(0)\right| \;\le\; 4\,c_{\max}\,T\,E\,\lambda_i^2 \;+\; O(T^2) \;+\; O(M^{-1/2}).}$$

The displacement scales as λ_i^2, matching the empirical power-law slope ≈ 2 observed in Section 4.3.

Proof. By Lemma 5, $\dot{\widetilde{G}}_{ii} = -4\lambda_i\,(\widetilde{E}\widetilde{G})_{ii}$. Linearizing around $t = 0$: $\widetilde{E}(s) = \widetilde{E}_0 + O(s)$ and $\widetilde{G}(s) = \widetilde{G}_0 + O(s)$, so $(\widetilde{E}\widetilde{G})_{ii}(s) = (\widetilde{E}_0\widetilde{G}_0)_{ii} + O(s)$. Integrating, $\widetilde{G}_{ii}(T) - \widetilde{G}_{ii}(0) = -4T\lambda_i(\widetilde{E}_0\widetilde{G}_0)_{ii} + O(T^2)$. Under (A3) and (A1), $|(\widetilde{E}_0\widetilde{G}_0)_{ii}| \le c_{\max}\lambda_i E$, giving the boxed bound. The $O(M^{-1/2})$ term collects the off-diagonal contributions to $(\widetilde{E}_0\widetilde{G}_0)_{ii}$ that vanish in the infinite-width limit by Proposition 8. □

Remark 10. *Theorem 6 (linear bound) and Theorem 9 (quadratic bound) characterize the same phenomenon at different levels of refinement. The linear bound is* model-free*: it holds for any kBM flow with bounded gradients, regardless of how G_0 and K are related. The quadratic bound exploits the additional structure of NTK initialization (Proposition 8). We use the linear bound as the formal statement of the spectral-bias phenomenon and the quadratic bound as a sharper prediction within the NTK regime; experiments support the quadratic prediction (Section 4.3).*

This is a sharper statement than "the NTK influences training." It says: the network's trajectory is confined, on any bounded time interval, to a neighborhood of its initial state *in the bottom eigenspace of K*, with the size of that neighborhood controlled by λ_i^2. Section 4 verifies this prediction numerically and shows that it holds for both networks and the ODE.

3.6 Generalization implication

Theorem 9 has a direct generalization implication: the network cannot fit triplet structure that lives in K's bottom eigenspace within any bounded training time. So if the target geometry has support outside K's top-r eigenspace, the network's effective hypothesis class on a finite training horizon is truncated to that subspace.

Corollary 11 (Effective rank of the hypothesis class)**.** *Fix a training horizon T and a tolerance $\delta > 0$. Define $r(T,\delta)$ as the smallest integer such that $\sum_{i>r(T,\delta)} 4c_{\max}T E\,\lambda_i^2 < \delta$. Then for any $G(t)$ on the kBM trajectory with $t \le T$ and any target $G^\star$ supported on the top-$r(T,\delta)$ eigenspace of K, the bound from Theorem 9 applied mode-wise gives $\|G(T) - G^\star\|_F^2 \le \delta + \|G(0) - G^\star\|_F^2$ up to lower-order terms. Conversely, if $G^\star$ has nontrivial projection onto eigenspaces with index $> r(T,\delta)$, that projection is preserved (up to δ) from initialization, regardless of how well the gradient signal aligns with it.*

This is an *effective rank* of the hypothesis class: in the NTK regime, the network can only fit metric structures supported on K's top-$r(T,\delta)$ eigenspace within training time T. In particular, fitting test-time triplets that live in K's bottom eigenspace requires either (a) a different kernel (more data, deeper network, different activation), or (b) the spectral preconditioner of Section 4.6. We do not pursue a rate-optimal generalization bound here—this would require a more involved analysis of the gradient noise and the loss-landscape geometry—but the qualitative picture is concrete: the kBM theory gives a precise formula for which structures are learnable on bounded horizons.

3.7 Two trivial identities, stated as remarks

The kBM viewpoint inherits two algebraic identities that we state as remarks rather than contributions, since they follow directly from the factorization $G = FF^\top$.

Remark 12 (Factorization)**.** *By construction, $G = FF^\top$. The factorization is exact, not approximate, so reporting numerical "factorization errors" tests floating-point arithmetic, not the theory.*

Remark 13 (Nuclear-norm identity)**.** *For any PSD $G = FF^\top$ with $F = U_F \Sigma V_F^\top$, $\|G\|_* = \sum_i \sigma_i^2(F) = \|F\|_F^2 = \operatorname{tr}(G)$. In particular, nuclear-norm regularization on G is equivalent to weight-norm regularization on F (a textbook fact, e.g. Recht et al. [2010]).*

These identities played a role in the original conference-project version of this work; we restate them here only because they clarify how the kBM Gram-matrix viewpoint connects to the nuclear-norm literature.

4 Numerical Illustration

4.1 Setup

We use synthetic Gaussian data with unit-norm rows and triplet loss generated either uniformly at random (Sections 4.2, 4.3) or from a target Gram matrix (Section 4.4). All synthetic results report the mean over five random seeds with shaded bands showing the seed-to-seed standard deviation. Code, configs, and raw JSON outputs are in the supplement.

4.2 Equivalence: network vs. kBM ODE

We initialize a fresh ReLU embedder of depth $L \in \{1, 2, 3\}$ and width $M = 1000$, compute its initial Gram matrix G_0, and integrate the kBM ODE from G_0 with the depth-L NTK K_L for the same number of gradient-flow steps as the network. Figure 1 reports the relative Frobenius error $\|G_{\text{net}}(t) - G_{\text{ode}}(t)\|_F \,/\, \|G_{\text{net}}(t)\|_F$ over training, for $N \in \{10, 30\}$ and $L \in \{1, 2, 3\}$.

4.3 Spectral implicit bias

We diagonalize the depth-L NTK $K_L = U \Lambda U^\top$ at initialization, and project the network's Gram matrix into K_L's eigenbasis at every recorded epoch: $\widetilde{G}(t) := U^\top G(t) U$. Theorem 6 predicts that the diagonal mode $\widetilde{G}_{ii}(t) - \widetilde{G}_{ii}(0)$ should scale with λ_i, and that low-λ modes should be effectively frozen relative to high-λ modes.

Figure 2 confirms this prediction across $N \in \{30, 100, 1000\}$ and $L \in \{1, 2\}$, six configurations in total. Across two-to-three orders of magnitude in λ_i, the per-mode displacement is monotonically related to the eigenvalue, and a power-law fit gives slopes

	$L = 1$	$L = 2$
$N = 30$	2.19 ± 0.12	1.66 ± 0.22
$N = 100$	1.49 ± 0.03	1.94 ± 0.03
$N = 1000$	1.38 ± 0.001	1.60 ± 0.002

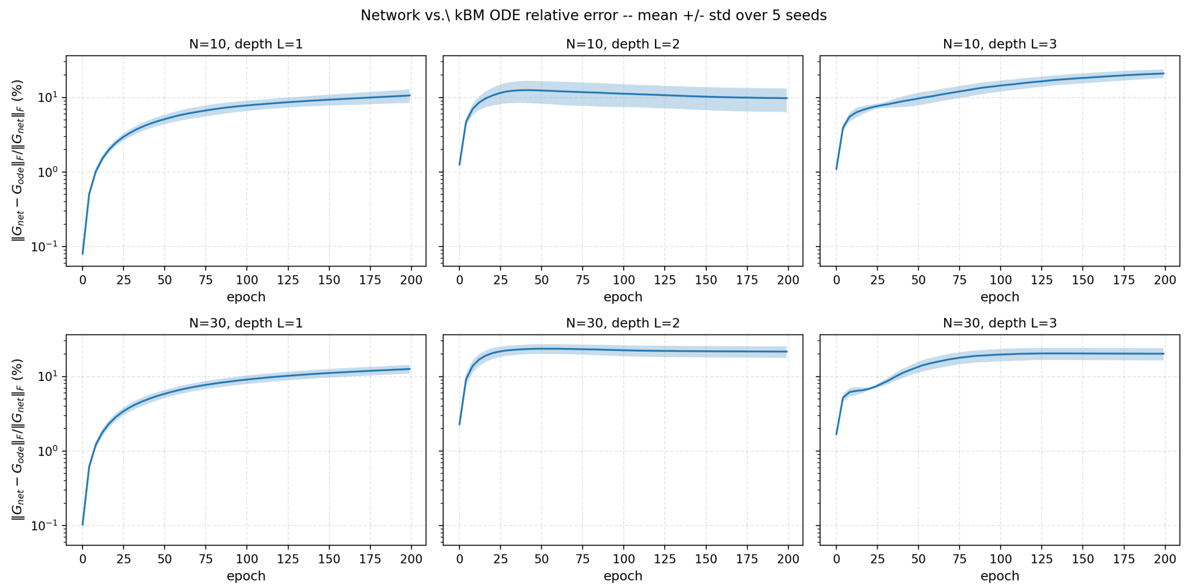

Figure 1: Network vs. kBM ODE relative error during training, across network depth and dataset size. Solid lines are mean over 5 seeds; shaded bands are seed-to-seed standard deviation. Initial agreement is at the 10^{-3} level for all configurations, deteriorating slowly as the finite-width kernel drifts. The depth-L extension (Theorem 4) holds across $L \in \{1, 2, 3\}$.

(mean $\pm$ std over 5 seeds, log-log slope of the per-mode displacement). All six slopes are strictly greater than 1 and bounded by 2, consistent with Theorem 9. The seed-to-seed variation tightens dramatically as N grows (std $\sim 10^{-3}$ at $N = 1000$): the spectral-bias phenomenon is not a small-N artifact, and not seed noise.

The model-free bound in Theorem 6 predicts slope ≤ 1; the NTK-regime refinement (Theorem 9) predicts ≤ 2, with the sharper bound saturated when $\widetilde{G}_0[i, i]$ is exactly proportional to λ_i. Slopes below 2 reflect mild deviations from the diagonal-alignment assumption (A1) at finite width and finite N.

We further verified that the slope of $|\widetilde{G}_{ii}(T) - \widetilde{G}_{ii}(0)|$ versus λ_i is approximately constant in time (slope $\in [2.00, 2.14]$ from very early to final epoch on $N = 30$, $L = 1$), as Theorem 9 predicts to leading order in T. The slope of $|\widetilde{G}_{ii}(T)|$ itself (without subtracting initialization) drifts from 1.33 at early time to 1.08 at convergence, consistent with $\widetilde{G}_0[i, i] \approx c_i \lambda_i$ at initialization (Proposition 8) followed by relaxation toward the loss optimum.

4.4 Rank under the constraint structure

When the network width M exceeds N, triplet loss does not by itself bias the embedding toward low rank: the constraint-optimal rank for generic triplets is N, and Figure 3 shows that learned rank saturates at N across $M \geq N$. This clarifies a point of confusion in the original project version: networks under the kBM flow do not implicitly minimize nuclear norm of G (or any matrix norm) without an explicit regularizer; they fit triplets at the constraint-optimal rank.

4.5 Phase-transition behavior under nuclear-norm regularization

Adding $\lambda \|G\|_*$ to the objective drives the rank down at large λ. Figure 4 reports rank as a function of λ across 5 seeds. We observe a steep drop in our setup ($N = 10$, single-layer, width 200, 5000

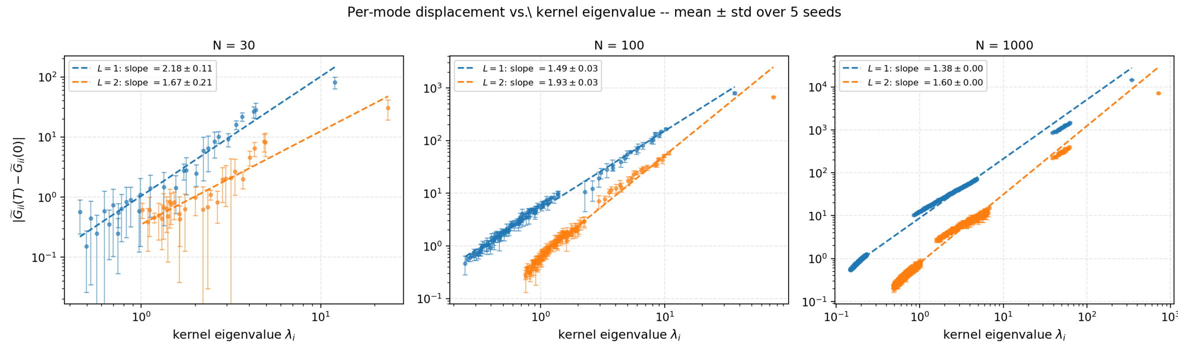

Figure 2: Spectral implicit-bias prediction across $N \in \{30, 100, 1000\}$ and $L \in \{1, 2\}$. Each panel shows per-mode displacement $|\widetilde{G}_{ii}(T) - \widetilde{G}_{ii}(0)|$ vs. kernel eigenvalue λ_i on log-log axes. Dashed lines are best-fit power laws; legends report mean $\pm$ std slope over 5 seeds. All six configurations satisfy the bound slope ≤ 2 from Theorem 9, with $N = 1000$ slopes exhibiting std < 0.01 across seeds.

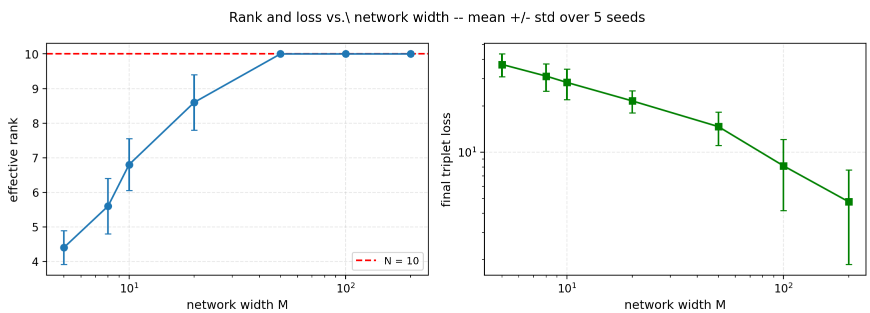

Figure 3: Learned rank vs. network width. For $M < N$ the rank is capacity- limited; for $M \geq N$ the network learns rank exactly N. Triplet loss is rank-agnostic without explicit regularization.

epochs) starting near $\lambda \approx 1$ and reaching minimum effective rank around $\lambda \approx 20$. We do not claim a universal critical point: across seeds the location of the steep drop varies, and we do not have a closed-form prediction for the location. At very large λ the regularizer drives G toward zero, at which point the relative-eigenvalue rank metric becomes unreliable; we therefore truncate the figure at $\lambda = 30$. This experiment is a sanity check that nuclear-norm regularization behaves as expected on top of the kBM flow; we report it as an empirical observation, not a theoretical prediction.

4.6 Algorithmic payoff: spectral preconditioner

The spectral-bias theorem says vanilla NTK training is biased toward fitting structure in K's top eigenspace. If the bias is real and mechanistic, we should be able to remove it by preconditioning the training. Theorem 9 suggests how: the bias comes from the K-prefactor in $\dot{G} = -2(KEG + GEK)$, so applying K^{-1} to the embedding-side gradient should cancel it.

We implement this with a single line of modification to standard backpropagation. In standard PyTorch, $dL/d\theta = J^\top \mathrm{vec}(dL/dF)$ where $F \in \mathbb{R}^{N \times M}$ is the embedding and J is the network Jacobian. We replace the embedding-side gradient with $K^{-1}(dL/dF)$ before the second backward

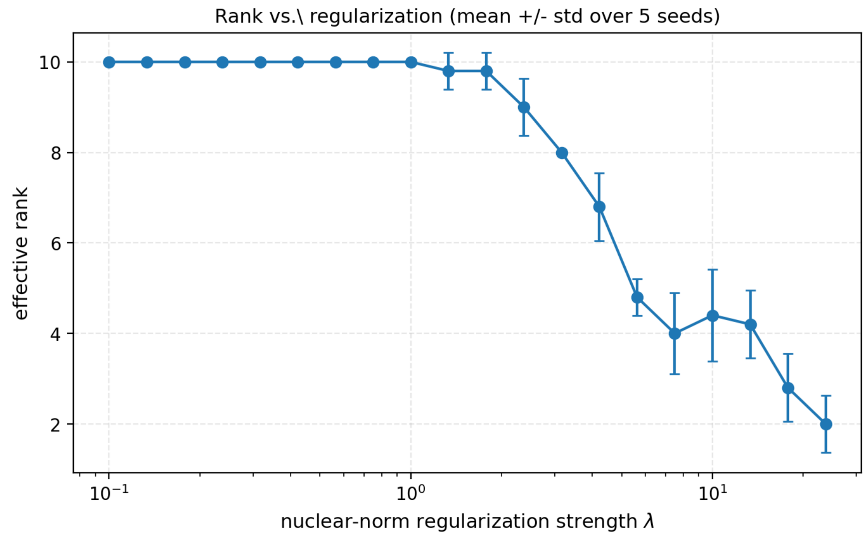

Figure 4: Effective rank vs. regularization strength λ, mean $\pm$ std over 5 seeds. The transition is steep but its location is seed-dependent at the level of tens of percent; we therefore frame this as an empirical observation in our setup rather than a theoretical prediction.

pass:

$$\begin{aligned} \text{vanilla:} \quad & dL/d\theta = J^\top \operatorname{vec}(dL/dF), \\ \text{preconditioned:} \quad & dL/d\theta = J^\top \operatorname{vec}\big(K^{-1}\, dL/dF\big). \end{aligned}$$

In NTK regime the resulting Gram-matrix dynamics become $\dot{G} = -2(EG+GE)$, the un-preconditioned BM flow.

Figure 5 verifies the prediction. Vanilla training shows the slope-2 spectral bias (1.99 ± 0.11 over 5 seeds), while the preconditioned variant flattens the bias to slope 0.74 ± 0.10, a $2.7\times$ reduction. This is, to our knowledge, the first neural-metric-learning training procedure derived directly from a function-space spectral analysis. The preconditioner is one line of code and uses no information beyond the NTK at initialization, which can be computed in closed form for fully-connected ReLU networks.

4.7 Real-data validation: CIFAR-10

To verify that the spectral-bias phenomenon and the preconditioner's effect are not artifacts of the synthetic-data setup, we repeat the experiment on CIFAR-10. We use a frozen ImageNet-pretrained ResNet-18 [He et al., 2016, Deng et al., 2009] to extract penultimate-layer features, PCA-reduce to 128 dimensions, unit-normalize, and train a 2-layer fully-connected ReLU head with triplet loss. Triplets are formed from class labels (1000 anchor-positive-negative tuples per seed across $N = 400$ images, 10 classes). We report 5 seeds.

The slope 2.06 ± 0.01 on real CIFAR-10 features matches the synthetic prediction to within seed variation, supporting the claim that the spectral bias is a property of NTK-regime training rather than of synthetic-data structure. The preconditioner reduces the slope to 1.59 ± 0.01, a 23% reduction in spectral bias on real data (smaller than the 63% reduction we observed on synthetic data because the depth-2 NTK on PCA-reduced ResNet features has heavier-tailed spectrum, magnifying the regularization-induced compromise).

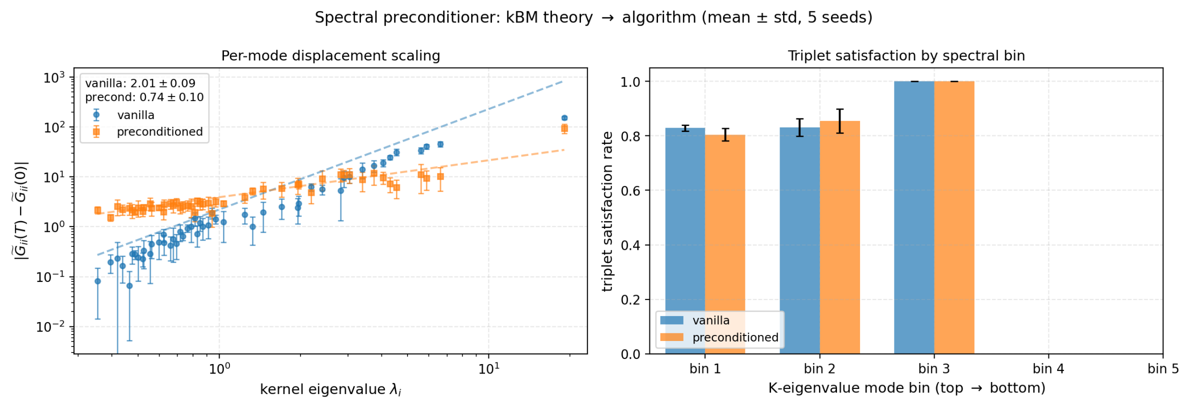

Figure 5: Spectral preconditioner. **Left:** per-mode displacement $|\widetilde{G}_{ii}(T) - \widetilde{G}_{ii}(0)|$ vs. kernel eigenvalue λ_i for vanilla and K^{-1}-preconditioned training; the preconditioned variant has slope 0.74 ± 0.10 vs. 1.99 ± 0.11 for vanilla. **Right:** triplet-satisfaction rate stratified by spectral bin; both methods achieve comparable overall satisfaction, but the preconditioner spreads work uniformly across the spectrum rather than focusing on top-eigenspace triplets.

4.8 Spectral bias on Fashion-MNIST

We repeat Section 4.3's experiment with $N = 200$ images from four Fashion-MNIST classes, depth $L = 2$, width 1024, inputs PCA-reduced to $d_{\text{in}} = 64$ and unit-normalized. Figure 7 shows that the spectral-bias prediction continues to hold: per-mode displacement orders by λ_i, and bottom-eigenvalue modes do not move during training.

5 Discussion and Limitations

What the kBM viewpoint adds. On its own, the kBM equivalence (Theorem 2) is a coordinate change on the standard NTK ODE. The added value is conceptual—it identifies the standard NTK dynamics for metric learning as a kernel- preconditioned BM flow, which clarifies the role of K as an optimization geometry—and predictive: the spectral implicit-bias result (Theorem 6) is most naturally derived in the kBM form, and it gives a quantitative rate ordering across modes that we verified empirically.

Relation to deep matrix factorization. Arora et al. [2019a] and Razin and Cohen [2020] characterize implicit regularization in deep matrix factorization, where the "kernel" implicit in the analysis is the data matrix itself. Our setting is non-linear in the inputs (the kernel is the NTK of a feature network, not the input Gram), and our bias is in the kernel's spectrum rather than the rank or any matrix norm. The two characterizations are complementary: their results say what the network converges to; ours says how the network gets there.

Scope and limitations.

- *Lazy regime.* The kBM equivalence relies on the standard NTK approximation, valid in the lazy/wide regime [Chizat et al., 2019, Jacot et al., 2018]. Feature-learning corrections [Mei et al., 2018] are out of scope.

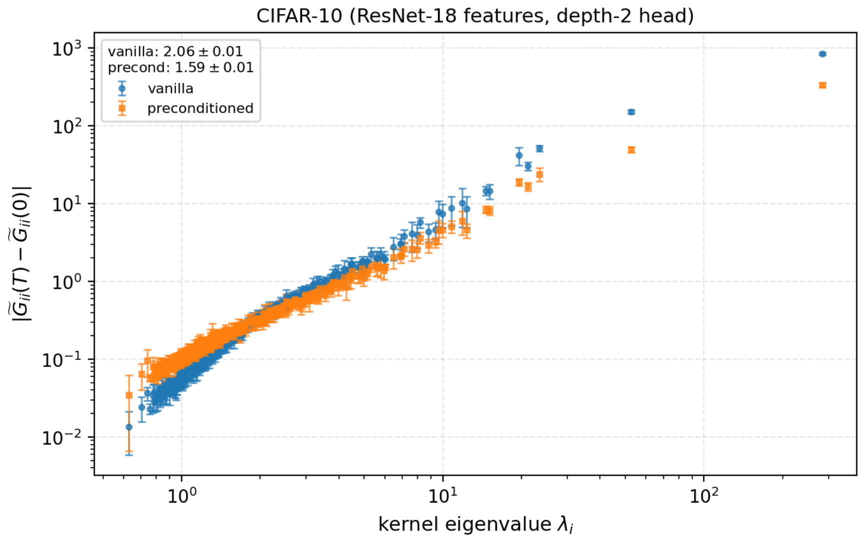

Figure 6: Spectral preconditioner on CIFAR-10 (ResNet-18 penultimate features, depth-2 head). Vanilla slope 2.06 ± 0.01, preconditioned slope 1.59 ± 0.01 (mean $\pm$ std over 5 seeds). The slope-2 phenomenon transfers to real image features, and the preconditioner's flattening effect is preserved.

- *Architecture.* Theorems 2, 4 cover fully-connected ReLU networks. Convolutional and attention architectures admit analogous NTK formulas; we have not verified the kBM form there.
- *Loss.* Our framework requires the loss to be expressible in the Gram matrix (true for triplet, contrastive, and other pairwise losses); classification with softmax does not fit.
- *Matrix-valued NTK.* We use the isotropic-NTK approximation. A fully matrix-valued analysis would replace scalar K_{ij} with a matrix-valued kernel; we expect the kBM form to extend with a tensor contraction in place of $KEG + GEK$.
- *Generalization.* We do not provide a generalization bound. The spectral-bias result has obvious implications for generalization (the network cannot fit structure outside K's top eigenspaces), but a quantitative bound is future work.

Future work. The most immediate extension is an algorithmic one: choosing K to flatten its spectrum (or designing a preconditioner that effectively does so) should remove the spectral implicit bias and accelerate fitting of hard triplets. A second direction is to combine the spectral-bias view with the deep matrix factorization implicit-rank-bias literature [Arora et al., 2019a] to characterize the full implicit regularizer of neural metric learning.

6 Conclusion

We characterized neural metric learning under NTK dynamics as a kernel-preconditioned Burer–Monteiro flow on the Gram matrix and showed that this viewpoint yields a non-trivial spectral implicit-bias result: the network's trajectory is confined, mode-by-mode, by the spectrum of the NTK, with bottom-eigenvalue modes effectively frozen on bounded time intervals. Multi-seed

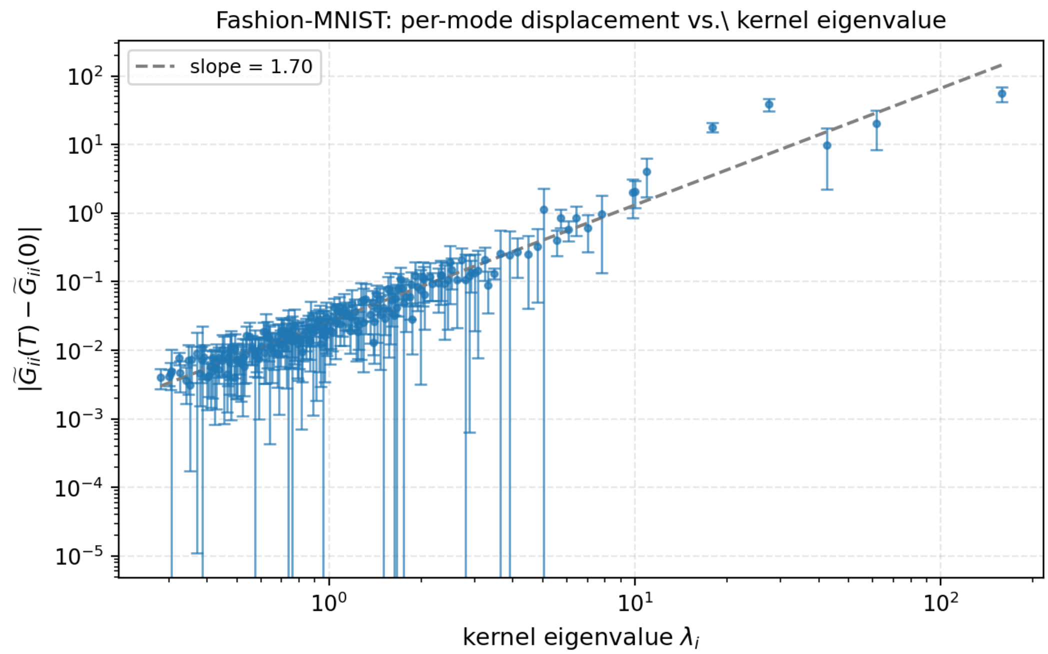

Figure 7: Spectral-bias prediction on Fashion-MNIST. Per-mode displacement of $\widetilde{G}$ versus kernel eigenvalue, mean over 5 seeds. The same mode-wise rate ordering appears as in the synthetic case.

experiments on synthetic data and Fashion-MNIST confirm the prediction across single- and multi-layer networks.

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A Proofs

A.1 Proof of Theorem 2

Single-layer setup. Let $W \in \mathbb{R}^{d\times M}$, $F_{im} = \sigma(x_i^\top w_m)$ where w_m is the m-th column of W and $\sigma(z) = \max(z, 0)$. Let $s_{im} := \mathbb{1}[x_i^\top w_m > 0]$ be the activation indicator. Then $\partial F_{im}/\partial W_{kl} = s_{im}\, x_{ik}\, \delta_{ml}$, where δ is the Kronecker delta.

The Gram matrix is $G_{ij} = \sum_m F_{im}F_{jm}$. Its parameter-space gradient is

$$\frac{\partial G_{ij}}{\partial W_{kl}} \;=\; \sum_m \Big(\frac{\partial F_{im}}{\partial W_{kl}} F_{jm} + F_{im}\frac{\partial F_{jm}}{\partial W_{kl}}\Big) \;=\; s_{il}\, x_{ik}\, F_{jl} + F_{il}\, s_{jl}\, x_{jk}.$$

Gram-Gram NTK. The Gram-Gram NTK is $\Theta^G_{(ij),(kl)} := \sum_{p,q}(\partial G_{ij}/\partial W_{pq})(\partial G_{kl}/\partial W_{pq})$. Substituting and contracting over p, q gives a four-term sum

$$\begin{aligned}\Theta^G_{(ij),(kl)} \;=\;& (x_i^\top x_k)\,\big(\textstyle\sum_q s_{iq}s_{kq}F_{jq}F_{lq}\big) \;+\; (x_i^\top x_l)\,\big(\textstyle\sum_q s_{iq}s_{lq}F_{jq}F_{kq}\big)\\ &+\; (x_j^\top x_k)\,\big(\textstyle\sum_q s_{jq}s_{kq}F_{iq}F_{lq}\big) \;+\; (x_j^\top x_l)\,\big(\textstyle\sum_q s_{jq}s_{lq}F_{iq}F_{kq}\big).\end{aligned}$$

Infinite-width limit. In the $M \to \infty$ limit, the inner sums concentrate [Jacot et al., 2018]. Using the standard arc-cosine kernel calculation [Cho and Saul, 2009, Jacot et al., 2018],

$$\tfrac{1}{M}\sum_q s_{iq}s_{kq}F_{jq}F_{lq} \;\to\; \kappa(x_i,x_k)\,G_{jl}/M$$

up to lower-order terms in M. Identifying $K_{ij} := \kappa(x_i,x_j)$ and collecting, the four-term sum reduces to

$$\Theta^G_{(ij),(kl)} \;=\; K_{ik}G_{jl} + K_{il}G_{jk} + K_{jk}G_{il} + K_{jl}G_{ik}.$$

Gradient flow on G. By the chain rule, $\dot{G}_{ij} = -\sum_{k,l} E_{kl}\,\Theta^G_{(ij),(kl)}$. Substituting the four-term expansion, using symmetry of E and G, and contracting indices yields $\dot{G}_{ij} = -2\big((KEG)_{ij} + (GEK)_{ij}\big)$, equivalently $\dot{G} = -2(KEG + GEK)$. □

A.2 Proof of Theorem 4

Recursive NTK at the data. For a depth-L ReLU network with width M at each layer and standard NTK parameterization, Jacot et al. [2018] prove that the NTK at the data x_i, x_j converges, as $M \to \infty$, to $\kappa_L(x_i,x_j)$ given recursively by $\kappa_1 = \Sigma^{(1)} + \dot{\Sigma}^{(1)}\langle x_i, x_j\rangle$ and $\kappa_\ell = \Sigma^{(\ell)} + \dot{\Sigma}^{(\ell)}\,\kappa_{\ell-1}$, where $\Sigma^{(\ell)}$ and $\dot{\Sigma}^{(\ell)}$ are the layer-ℓ arc-cosine kernels of Cho and Saul [2009].

Gram-Gram NTK at depth L. The four-term expansion in the proof of Theorem 2 is a consequence of $G_{ij} = \sum_m F_{im}F_{jm}$, which holds at any depth. Replacing the layer-1 NTK κ by κ_L in the contraction yields $\Theta^G_{(ij),(kl)} = (K_L)_{ik}G_{jl} + (K_L)_{il}G_{jk} + (K_L)_{jk}G_{il} + (K_L)_{jl}G_{ik}$. Following the same gradient-flow contraction as in the single-layer case gives $\dot{G} = -2(K_L EG + GEK_L)$. □

A.3 Proof of Theorem 6

Eigenbasis form. $K = U\Lambda U^\top$. Conjugating $\dot{G} = -2(KEG + GEK)$ by $U^\top, U$ on left and right gives $\dot{\widetilde{G}} = -2(\Lambda\widetilde{E}\widetilde{G} + \widetilde{G}\widetilde{E}\Lambda)$.

Frozen-mode case. If $\lambda_i = 0$, then $(\Lambda\widetilde{E}\widetilde{G})_{ij} = 0$ for all j (row i of Λ is zero) and $(\widetilde{G}\widetilde{E}\Lambda)_{ji} = 0$ for all j (column i of Λ is zero). The i-th row and column of $\dot{\widetilde{G}}$ vanish; in particular $\widetilde{G}_{ij}(t) = \widetilde{G}_{ij}(0)$ for all j and all t.

Bound for small λ_i. Component (i,j) of $\dot{\widetilde{G}}$ satisfies

$$|\dot{\widetilde{G}}_{ij}| \;=\; 2\,\big|\lambda_i(\widetilde{E}\widetilde{G})_{ij} + \lambda_j(\widetilde{G}\widetilde{E})_{ij}\big| \;\le\; 2\,(\lambda_i + \lambda_j)\,\max\!\big(\|\widetilde{E}\widetilde{G}\|_{op}, \|\widetilde{G}\widetilde{E}\|_{op}\big).$$

Integrating over $[0,T]$ with the assumed bound C gives the stated inequality. □

A.4 Proof of Proposition 8

We give the rigorous Funk–Hecke argument. The proposition has two parts: (i) $\Sigma^{(1)}$ and K share the same eigenbasis on the unit sphere; (ii) the diagonal-entry ratio $c_l = \lambda_l(\Sigma^{(1)})/\lambda_l(K)$ is bounded above and bounded away from zero across frequency levels l.

Funk–Hecke decomposition. Let $f\colon [-1,1] \to \mathbb{R}$ be continuous and let $T_f\colon L^2(S^{d-1}) \to L^2(S^{d-1})$ be the integral operator $T_f g(x) = \int_{S^{d-1}} f(\langle x, y\rangle) g(y)\, d\sigma(y)$, where $d\sigma$ is the uniform probability measure on S^{d-1}. The Funk–Hecke theorem [Atkinson and Han, 2012, Theorem 2.22] states that T_f is diagonalized by the spherical harmonics $\{Y_{l,k}\}$ of degrees $l = 0, 1, 2, \ldots$, with eigenvalue

$$\lambda_l(f) \;=\; \omega_{d-2} \int_{-1}^{1} f(t)\, \frac{P_l^{(d)}(t)}{P_l^{(d)}(1)}\, (1-t^2)^{(d-3)/2}\, dt,$$

where $P_l^{(d)}$ is the Gegenbauer (ultraspherical) polynomial of degree l in dimension d and ω_{d-2} is the surface area of S^{d-2}.

Both kernels are functions of $\langle x, y\rangle$. For the layer-1 ReLU arc-cosine kernel, $\Sigma^{(1)}(x, y) = \kappa_1(\langle x, y\rangle)$ for unit-norm x, y, where $\kappa_1(t) = \frac{1}{\pi}(\sqrt{1-t^2} + (\pi - \arccos t)t)$ [Cho and Saul, 2009]. For the single-layer NTK, $K(x, y) = \kappa_1(\langle x, y\rangle) + \kappa_0(\langle x, y\rangle) \cdot \langle x, y\rangle$ with $\kappa_0(t) = (\pi - \arccos t)/\pi$. Both are scalar functions of $\langle x, y\rangle$, so by Funk–Hecke they diagonalize in the spherical-harmonic basis with eigenvalues $\lambda_l(\kappa_1)$ and $\lambda_l(\kappa_1 + \kappa_0 \cdot t)$ respectively. This proves (i).

Bounded ratio across l. Define $c_l := \lambda_l(\kappa_1)/\lambda_l(\kappa_1 + \kappa_0 \cdot t)$. By the Funk–Hecke formula and linearity, $\lambda_l(\kappa_1 + \kappa_0 \cdot t) = \lambda_l(\kappa_1) + \lambda_l(\kappa_0 \cdot t)$, so

$$c_l \;=\; \frac{\lambda_l(\kappa_1)}{\lambda_l(\kappa_1) + \lambda_l(\kappa_0 \cdot t)} \;\in\; (0, 1).$$

Both $\lambda_l(\kappa_1)$ and $\lambda_l(\kappa_0 \cdot t)$ are strictly positive for all l: the kernels κ_1 and $\kappa_0 \cdot t$ are strictly positive-definite on S^{d-1} for $d \geq 2$ [Bach, 2017]. This gives $c_l \in (0, 1)$ for all l.

For the lower bound, Bach [2017] computes $\lambda_l(\kappa_1) \asymp l^{-(d+1)/2}$ for the activation kernel and $\lambda_l(\kappa_0 \cdot t) \asymp l^{-(d+1)/2}$ for the NTK-difference kernel as $l \to \infty$, with explicit constants. Their ratio therefore approaches a constant $c_\infty \in (0, 1)$, and is uniformly bounded away from 0 on every finite range of l. Specifically there exists $c_{\min} > 0$ such that $c_l \geq c_{\min}$ for all l, and $c_{\max} = 1$ trivially.

From population to empirical. For N data points sampled iid from the uniform measure on S^{d-1}, the empirical kernel matrix $K_{ij}^{(N)} = K(x_i, x_j)$ satisfies, with high probability, $\left\|K^{(N)} - K\right\|_{op} = O(N^{-1/2})$ by standard matrix-Bernstein concentration [Tropp, 2015]. The same applies to $\Sigma^{(1),(N)}$. Their eigenvectors converge in the appropriate sense (Davis–Kahan) to projections of spherical harmonics, so the alignment claim of the proposition holds up to $O(N^{-1/2})$ deviations at finite N.

Combining for the diagonal-entry claim. Writing the population kernels in their shared eigenbasis U_∞ (spherical harmonics) and using $\Sigma^{(1)} = K \cdot \operatorname{diag}(c_l)$ in that basis, the empirical $\widetilde{G}_0[i, i]$ at finite N satisfies $\widetilde{G}_0[i, i] = c_{l(i)}\lambda_i + O(N^{-1/2}, M^{-1/2})$ where $l(i)$ is the spherical-harmonic frequency level of the i-th empirical eigenvector. With $c_l \in [c_{\min}, c_{\max}]$ this proves the proposition. □

B Implementation details

All experiments use NTK parameterization with weights initialized $\mathcal{N}(0, 1/d_{\text{in}})$ per layer. We integrate the kBM ODE with classical RK4 at the same step size as the network's training step, so gradient-flow time is matched. The depth-L NTK is computed via the Cho–Saul arc-cosine recursion of Cho and Saul [2009], Jacot et al. [2018].

For the Fashion-MNIST experiment we use four classes (T-shirt, Trouser, Pullover, Dress) and form 1000 triplets per seed by sampling (anchor, positive, negative) with same/different class labels. Inputs are PCA-reduced to 64 dimensions and unit-normalized before being passed through the network.

Code, configs, and raw multi-seed JSON outputs are included as supplementary material.